

User-defined keyboard types

Ukelele, as of version 3.6, supports user-defined keyboard types, that is, arrangements of physical keys as displayed on screen. The file type for this is the Apple property list, which is a particular form of XML. At present, the only way to generate the file is by hand in an editor such as Xcode, BBEdit or similar, with copying from another keyboard type as the only short-cut. It's possible that one day there will be an app for doing this, but no promises!

This document describes the format of the XML file and the way it is used to build a screen image for Ukelele. For reference, the built-in keyboard types are stored in a property list within the Ukelele app bundle, at Ukelele.app/Contents/Resources/KeyboardTypes.plist.

File format

The file must be a property list, that is, an XML file with the file extension plist. The file must conform to the property list specification, so have headers that look like this:

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
```

The body of the definition is a dictionary, so begins with:

```
<dict>
```

and ends with:

```
</dict>
```

The whole file ends with:

```
</plist>
```

The contents of the dictionary are keyboard type definitions, each of which consists of a key which is an integer and a dictionary containing the definition. So the format will be:

```
<key>1234</key>
<dict>
```

```
...
```

```
</dict>
```

The number you use for the key is an arbitrary positive integer, and is mainly used to satisfy the requirements of the property list to have distinct item keys. When Ukelele imports the definition, the key is used to generate an internal key which is unique, with user-defined keyboard types having a key number over 10,000, so as to avoid conflict with actual hardware IDs, which have numbers under 1,000.

The dictionary for the definition has the following entries:

Key	Content
name	A string giving the name that will appear in the list of keyboard types in the Keyboard Type dialog.
description	A string giving the longer text that shows in the bottom of the Keyboard Type dialog.
coding	A string, one of ANSI, ISO or JIS.
keys	An array of key definitions.

The coding is not actually used to display the keyboard, so it is arbitrary which one you use. It appears in the Keyboard Type dialog, but is not otherwise used. The reason it is there is because it is using the same format to read the built-in Apple keyboard types, for which it does actually matter.

Each key definition is a dictionary consisting of the following entries:

Key	Content
KeyCode	An integer, which is the key code.
Rect1	An array of four real numbers defining the rectangle.
Rect2	An optional array of four real numbers defining the rectangle.

The keys are an array because there are usually multiple keys with the same key code, namely the modifier keys, so that there are two shift keys, two option keys, etc.

Each rectangle array has four numbers, which are usually integers although it is possible to be non-integer real numbers. The first two numbers are the origin point of the rectangle, that is, its lower left corner, the first number being the horizontal position (the x axis coordinate) and the second the vertical position (the y axis coordinate). The third number is the width of the rectangle, and the fourth is the height. A sample rectangle entry looks like this:

```
<array>
  <real>522</real>
  <real>62</real>
  <real>58</real>
  <real>38</real>
</array>
```

Most keys are defined with a single rectangle, but you can also use two rectangles for L-shaped keys. It does not matter which order you use for the two rectangles, but they should connect, so one edge continues from one rectangle to the other, and one edge ends part-way along one of the other rectangle's edges.

Making a design into the right format

The way that the keyboard definition works is that there is a large rectangle with the origin at (0, 0), and contains all the rectangles listed in the definition, and extending two points beyond the furthest horizontal and vertical rectangle corners. All rectangles are aligned to the grid, so that you can't have keys at an angle or shapes other than a simple rectangle or an L-shape defined by two rectangles.

The way that the default size is set up is that a square key is 38 × 38 with a one point margin on all sides, so essentially on a 40 × 40 grid. The bottom left key has its origin at (2, 2).

The size of the non-square keys is variable. Usually they will be 38 points high and align with the other keys on the same row. Small keys, like arrow keys or function keys on some keyboards, are 18 points high with a one point margin, so 40 × 20. The non-square keys like tab, shift, backspace, etc, have varying widths depending on the keyboard, so you may need to experiment to see how wide you need them to be.

If there is an L-shaped key, it will almost certainly be the return key, and there is quite a variety of ways this can look. The trick is mainly to take into account the margin to get the rectangles to meet. So if the key is like an upside down L, then the rectangle for the lower part needs to be taller by two points, so often 40 points rather than 38.

Using keyboard definitions

When you have a keyboard definition in a property list file, you can import it from the Keyboard Type dialog. The list should automatically update so that your new keyboard definition is available, down towards the bottom of the list. It is saved as part of the Ukelele preferences, so should remain available, and can be selected as the type to use by default.

If you try a keyboard definition and it doesn't look right, then you either need to add it again and have both versions, or you can delete all the user-defined keyboard types and add the good ones (if any) again. This is done from the Preferences window, at the bottom right. At present, there is no way to delete them selectively, so you want to keep the original files around for any keyboard types that you want to use.

If there is an error reading the file, you will get an alert that says that some keyboard types were not valid. There will be some more information in the logs, which you can look at with the Console utility provided by Apple (Applications/Utilities/Console.app). Using Console is not as easy as it used to be, so you will most likely want to open it before trying the import, and filtering to messages containing Ukelele.

Key codes

The table lists key codes from 0 to 127, along with the corresponding key on the default US keyboard layout. Note that several codes are simply unknown, as they do not correspond to any key I know. Also note that there are key codes for right shift, right option and right control. You don't want to use those, but use the shift key code for both shift keys.

If you have keys not listed here, you can work out the key code by using the Inspector window in Ukelele to show the key code for the key just pressed.

Key code	Key output or use
0	A
1	S
2	D
3	F
4	H
5	G
6	Z
7	X
8	C
9	V
10	Section mark §
11	B
12	Q
13	W
14	E
15	R
16	Y
17	T
18	1
19	2
20	3
21	4
22	6
23	5
24	Equals =

25	9
26	7
27	Hyphen -
28	8
29	0
30	Close square bracket]
31	O
32	U
33	Open square bracket [
34	I
35	P
36	Return
37	L
38	J
39	Single quote '
40	K
41	Semicolon ;
42	Backslash \
43	Comma ,
44	Slash /
45	N
46	M
47	Dot .
48	Tab
49	Space
50	Back tick `
51	Delete/backspace
52	<i>Unknown</i>
53	Escape
54	<i>Unknown</i>
55	Command
56	Shift
57	Caps lock
58	Option
59	Control
60	Right shift
61	Right option
62	Right control
63	Fn
64	F17
65	Keypad .
66	<i>Unknown</i>
67	Keypad *
68	<i>Unknown</i>

69	Keypad +
70	<i>Unknown</i>
71	Keypad Clear
72	<i>Unknown</i>
73	<i>Unknown</i>
74	<i>Unknown</i>
75	Keypad /
76	Keypad Enter
77	<i>Unknown</i>
78	Keypad -
79	F18
80	F19
81	Keypad =
82	Keypad 0
83	Keypad 1
84	Keypad 2
85	Keypad 3
86	Keypad 4
87	Keypad 5
88	Keypad 6
89	Keypad 7
90	<i>Unknown</i>
91	Keypad 8
92	Keypad 9
93	Yen (on JIS keyboard)
94	<i>Unknown</i>
95	<i>Unknown</i>
96	F5
97	F6
98	F7
99	F3
100	F8
101	F9
102	Japanese conversion key 英数
103	F11
104	Japanese conversion key カナ
105	F13
106	F16
107	F14
108	<i>Unknown</i>
109	F10
110	Contextual Menu (Windows keyboards)
111	F12

112	<i>Unknown</i>
113	F15
114	Help
115	Home
116	Page up
117	Forward delete
118	F4
119	End
120	F2
121	Page down
122	F1
123	Left arrow
124	Right arrow
125	Down arrow
126	Up arrow
127	<i>Unknown</i>